

NANYANG JUNIOR COLLEGE
PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9744/04

Paper 4 Practical

27 August 2018

Candidates answer on the Question Paper

Additional Materials: As listed in the Confidential Instructions

2 hour 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper

Shift
Laboratory

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
Total	

Answer **all** the questions.

- 1** You are required to estimate the water potential of potato tissue.

When a piece of potato is put into a sucrose solution, as shown in Fig. 1.1, water will move by osmosis into and out of the potato cells. The net direction of movement of water depends on the difference in water potential between the potato cells and the sucrose solution.

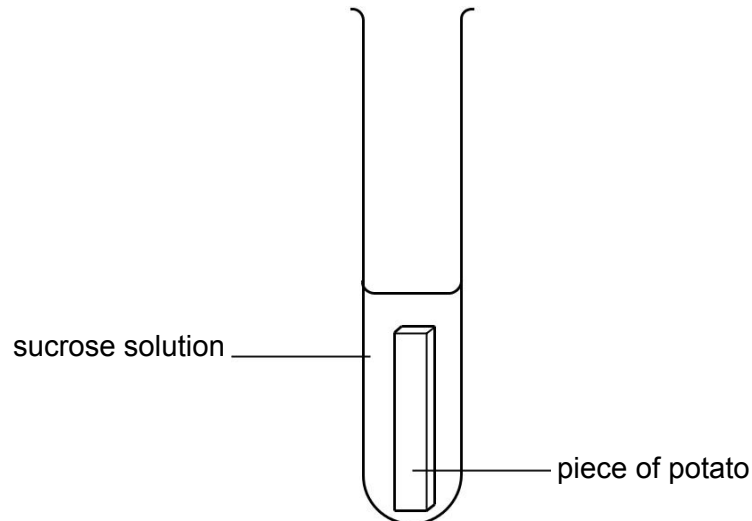


Fig. 1.1

- (a)** **(i)** Complete the sentences by using **two** of the following words.

gain

less

lose

more

If the potato cells water then the sucrose solution will become more dilute.

This will change the sucrose solution so that it becomes dense.

[1]

A piece of potato is left in a sucrose solution for 15 minutes, to allow time for osmosis to take place.

The concentration of the sucrose solution after 15 minutes may be different from the original concentration.

After 15 minutes a blue dye is added to the sucrose solution to make a blue solution. The blue dye does not affect the concentration of the sucrose solution.

A small volume of this blue solution is then released into a known concentration of sucrose solution as shown in Fig. 1.2.

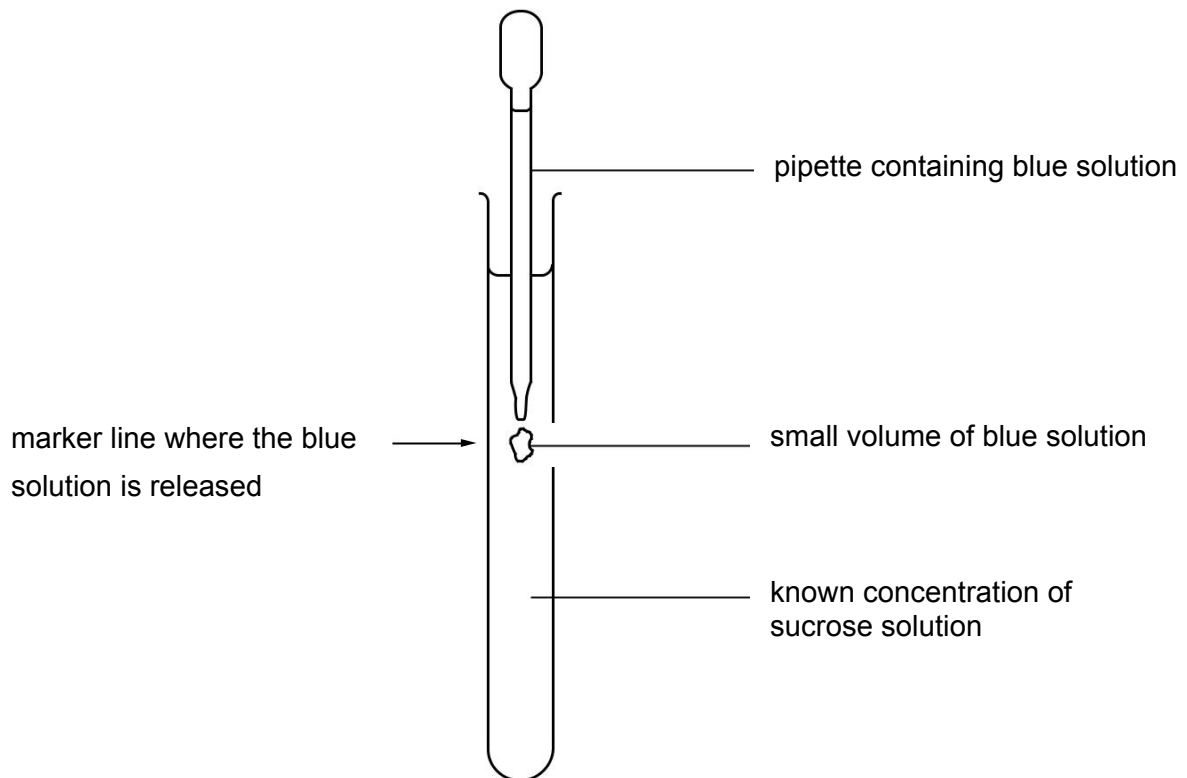


Fig. 1.2

The pipette is removed immediately after the blue solution is released.

The blue solution may move up, move down or remain at the same level depending on the difference in the concentration between the blue solution and the known concentration of sucrose solution.

(ii) Think about how the blue solution will move in the known concentration of sucrose in the test-tubes shown in Fig. 1.3.

blue solution is
less concentrated
than the known
sucrose solution



blue solution is the
same concentration
as the known
sucrose solution



blue solution is
more concentrated
than the known
sucrose solution

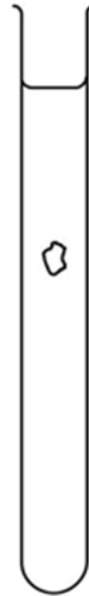


Fig. 1.3

Complete Fig. 1.3 by drawing an arrow on each test-tube, using the key, to show how you expect the blue solution to move.

key:

↑

blue solution moves up

↓

blue solution moves down

→

blue solution remains at the same level

[1]

(b) You are required to investigate osmosis in potato tissue so that you can estimate the water potential of the potato cells.

You are provided with:

labelled	contents	hazard	volume / cm ³
S	1.00 mol dm ⁻³ sucrose solution	none	200
W	distilled water	none	200

labelled	details
P	7 pieces of potato, cross sectional area 1.5 cm × 1.0 cm of variable length

You are required to make simple dilutions of the 1.00 mol dm⁻³ sucrose solution, **S**, which reduces the concentration between each dilution.

You will need to prepare 40 cm³ of each concentration.

Decide which concentrations of sucrose solution to prepare using simple dilutions of **S**.

(i) Complete Table 1.1 to show how you will prepare the other concentrations.

Table 1.1

final concentration of sucrose solution / mol dm ⁻³	volume of S / cm ³	volume of distilled water, W / cm ³
1.00	40	0

[3]

Read step 1 to step 11 before proceeding.

Proceed as follows:

1. Prepare all the concentrations of sucrose solution as shown in Table 1.1 in the beakers provided.
2. Measure 4.5 cm from the bottom of each **large** test-tube and put a mark, as shown in Fig. 1.4.

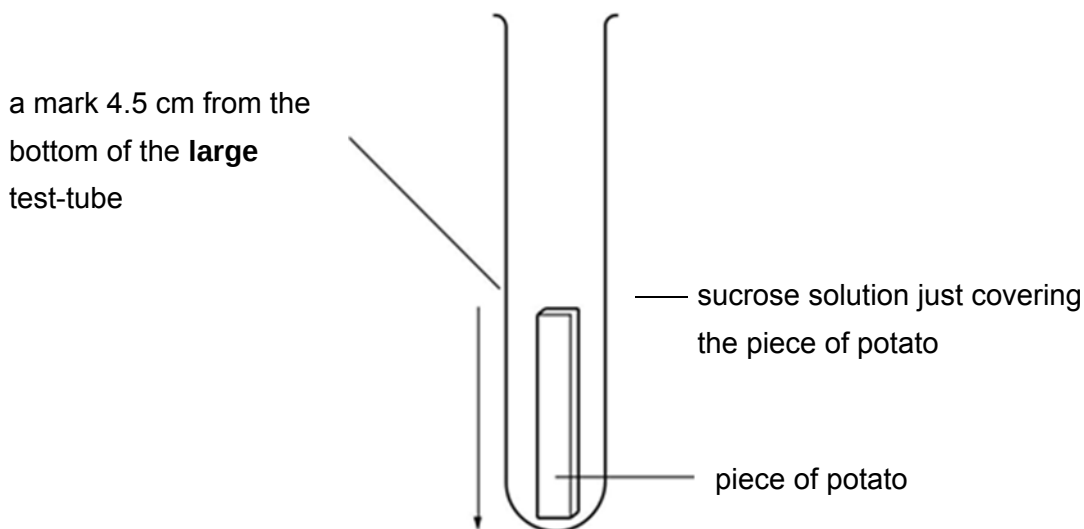


Fig. 1.4

- (ii) Decide on the length of each piece of potato you will use, so that the volume of sucrose solution just covers the piece of potato, as shown in Fig. 1.4.

State the length of the pieces of potato you will use.

length = [1]

3. Cut each piece of potato with a cross-sectional area of $1.5 \text{ cm} \times 1 \text{ cm}$, and the length in step 2 part (ii). Cut enough pieces of potato to put into the sucrose concentrations that you prepared in step 1.
4. Put the pieces of potato on a paper towel to remove any excess fluid.
5. Put one piece of potato into each of the large test-tubes from step 2.
6. Put 1.00 mol dm^{-3} sucrose solution, **S**, into a large test-tube up to the mark made in step 2.
7. Repeat step 6 with each of the other concentrations of sucrose solution.
8. Start timing and leave for 15 minutes.

While you are waiting for 15 minutes continue with step 9 to step 11 and continue with the other questions.

9. Measure 6 cm from the **top** of each of the **small** test-tubes and put a mark, as shown in Fig. 1.5.

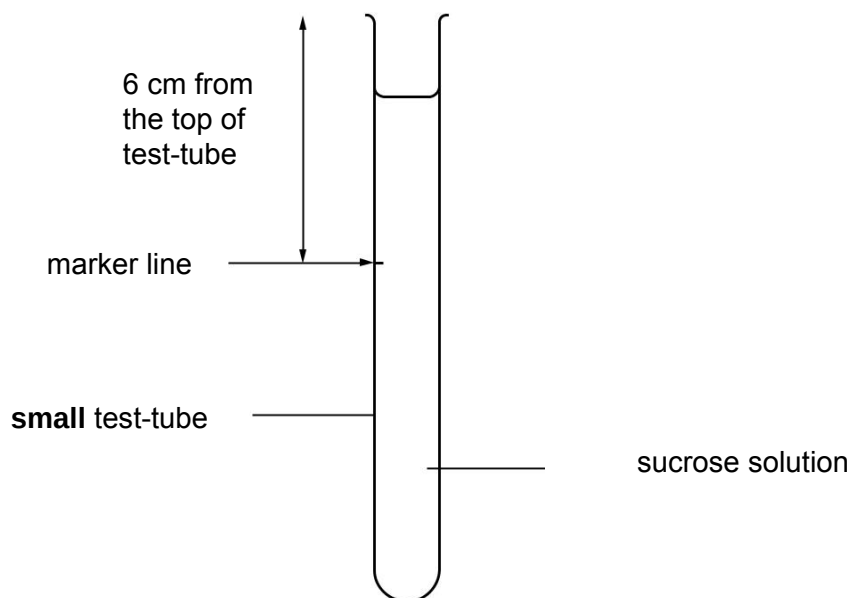


Fig. 1.5

10. Put 25 cm^3 of 1.00 mol dm^{-3} sucrose solution, **S**, into one of the **small** test-tubes from step 9.
11. Repeat step 10 with each of the other sucrose solutions that you prepared in step 1.

You are provided with:

labelled	contents	hazard	volume / cm^3
M	methylene blue solution	none	15

12. After leaving the pieces of potato for 15 minutes, put a **big** drop of **M** into the **large** test-tube containing 1.00 mol dm^{-3} sucrose solution, **S**, and a piece of potato.
13. Gently shake the **large** test-tube to mix **M** with the sucrose solution.
14. Repeat step 12 and step 13 with each of the other concentrations of sucrose solution.

Read step 15 to step 19 before proceeding.

15. Use a pipette to remove a sample of the blue solution from the **large** test-tube containing 1.00 mol dm^{-3} sucrose solution, **S**.

*You will now use the **small** test-tubes as in Fig. 1.5.*

16. Put the end of the pipette into the **small** test-tube containing 1.00 mol dm^{-3} concentration of sucrose solution, **S**. This should be level with the marker line on the test-tube as shown in Fig. 1.2 on page 3.
17. Release a small volume of the blue solution, then immediately remove the pipette from the test-tube.
It is possible to repeat step 17 without having to replace this sucrose solution.
18. Immediately observe the direction **and** the speed of movement of the blue solution. *You are **not** required to measure the speed.*

Record these observations in **(b)(iii)**.

19. Repeat step 15 to step 18 using the other concentrations of sucrose solution. Make sure that the small volume of the blue solution from the **large** test-tube is put into the **small** test-tube labelled with the same concentration.

- (iii)** Record your observations of direction and speed of movement in an appropriate table below.

- (iv) Using your results in (b)(iii) estimate the concentration of sucrose solution with a water potential equal to the water potential of the potato tissue.

.....
.....[1]

- (v) Identify **one** significant source of error in this investigation.

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.....
.....[1]

- (vi) Describe how you would use this procedure to produce a more accurate estimate of the concentration of sucrose solution with a water potential equal to the water potential of the potato tissue.

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.....
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.....
.....[3]

- (vii) Describe the movement of water when the concentration of the sucrose solution surrounding the piece of potato has a water potential equal to the water potential in the potato tissue.

.....
.....[1]

- (viii) Using the same procedure a student observed that the blue solution stayed in the same position (did not move up or down) in a concentration of sucrose solution of 0.3 mol dm^{-3} .

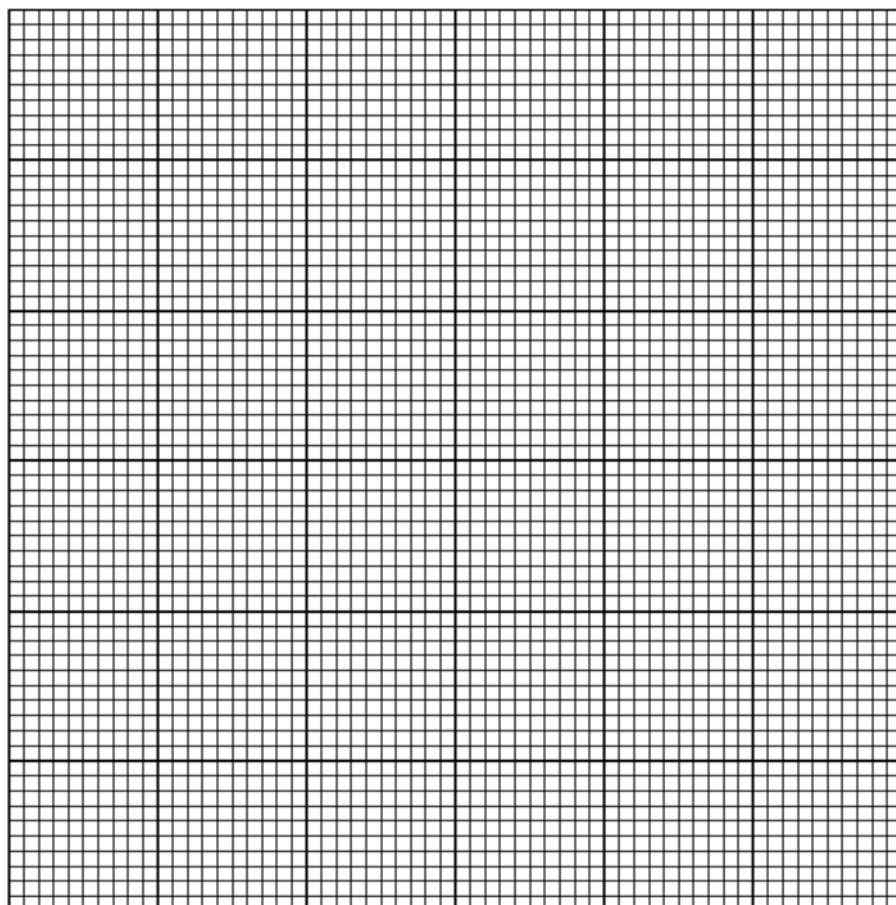
Table 1 shows the relationship between the concentration of sucrose solution and the water potential of the sucrose solution.

Table 1

Concentration of sucrose solution / mol dm^{-3}	Water potential / $\text{k Pa} \times 10^2$
0.2	26.0
0.4	12.0
0.6	5.0
0.8	2.5
1.0	1.5

Plot a graph using the data in **Table 1**.

[4]



[Total: 21]

- 2 You are provided with a quantity of vitamin C solution and a dye called DCPIP.

You are also provided with three test-tubes containing respectively lemon juice, orange juice, grapefruit juice, and labelled as such. These juices contain natural vitamin C and the dye DCPIP can be used to determine the concentration of this vitamin in the juices.

Proceed as follows:

- 1 Into a clean test-tube, transfer a quantity of the dye DCPIP to a depth of 0.5 cm. Take note its colour.
- 2 Fill a teat pipette with vitamin C solution. Add one drop of vitamin C solution to the DCPIP solution in the test-tube and shake gently. Continue to add the drops, counting the number of drops which are needed to bring about a colour change. Shake gently after each drop, refilling the pipette if necessary.
- 3 Record the initial colour of DCPIP (from step 1) and the first colour change after vitamin C is added as well as the number of drops counted to bring about this colour change in a suitable table.
- 4 After the first colour change, continue adding drops of vitamin C and counting the drops until the DCPIP solution becomes colourless/or consistent pale yellow. (Ignore any coloured granules that might form.). Record the number of drops counted in the same table from step 3.
- 5 Repeat steps 1 to 4 adequately to obtain enough data for analysis, cleaning all apparatus before use.
- 6 Place the DCPIP solution into each of three clean test-tubes to a depth of 0.5 cm. (The amount of DCPIP solution must be exactly the same in each of the tubes). Label the tubes A, B and C.
- 7 Fill a clean teat pipette with lemon juice and drop by drop add this to the contents of tube **A**, shaking the tube gently after each drop. Count the number of drops needed to turn the DCPIP solution colourless. Repeat this step adequately to obtain enough data for analysis.
- 8 Repeat the step 7 with orange juice and grapefruit juice, using a clean pipette each time to add the juice to the DCPIP solution in tubes **B** and **C** respectively.

9 Record the results for the three juices in an appropriate table. [4]

10 What conclusions can you draw from your results?

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.....[3]

- 11 Comment on the main source(s) of error and the limitations of the measurements or experimental procedure.

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.....[3]

- 12 What improvements could you make to the experimental procedures to overcome these sources of error?

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.....[2]

- 13 In this practical, you are required to investigate the effect of different wavelength of light on a leaf extract using a solution of the dye DCPIP (dichlorophenol indophenol). DCPIP turns from blue to colourless when it is reduced. The leaf extract will be prepared by you.

You may select from the following apparatus and use appropriate additional apparatus:

- Petri dish
- 6 melting point capillary tubes – 10 cm long checked to make sure both ends are open
- Plastic specimen tube (minimum 3 cm x 1 cm)
- Glass rod that fits the specimen tube
- White tile
- Sharp knife or scalpel
- One sheet of Aluminium foil, approximately 5 cm x 5 cm for capillary tube
- One sheet of Aluminium foil, approximately 12 cm x 12 cm to cover the Petri dish completely
- Bench lamp with 60W bulb
- Stopwatch
- 5-cm³ syringe
- Dropper pipette
- Paper towels
- Distilled water in a wash bottle
- 20 cm³ buffer solution dispensed in a Styrofoam cup containing crushed ice
- 10 cm³ DCPIP solution dispensed in a Styrofoam cup containing crushed ice
- 40 cm² fresh leaf (spinach), roughly 10 cm x 4 cm
- 2 cm x 5 cm green and red transparency paper

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of result tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[9]

[illegible]

9744 / H2 Biology / 04

- 3 (a) **M1** is a slide of a stained transverse section through a plant root.

You are not expected to be familiar with this specimen.

Use a sharp pencil for drawing.

- (i) Draw a large plan diagram of a quarter of the root on **M1**, shown by the shaded area in Fig. 3.1. A plan diagram only shows the arrangement of the different types of tissues. Individual cells must **not** be drawn in plan diagrams.

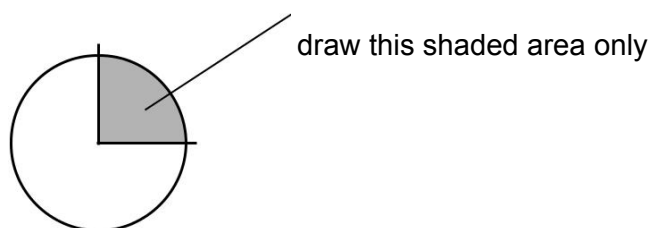


Fig. 3.1

Your drawing should show the correct shape and proportions of the different tissues.

- (ii) Observe the central tissue in the root on **M1**.

Select **one** large xylem vessel and **three** adjacent (touching) smaller xylem vessels from the tissue at the centre of the root. The large xylem vessel must touch each of the three smaller xylem vessels.

Make a large drawing of this group of **four** xylem vessels.

[4]

- (b) Fig. 3.2 is a photomicrograph of a stained transverse section of a different root. You are not expected to be familiar with this specimen.

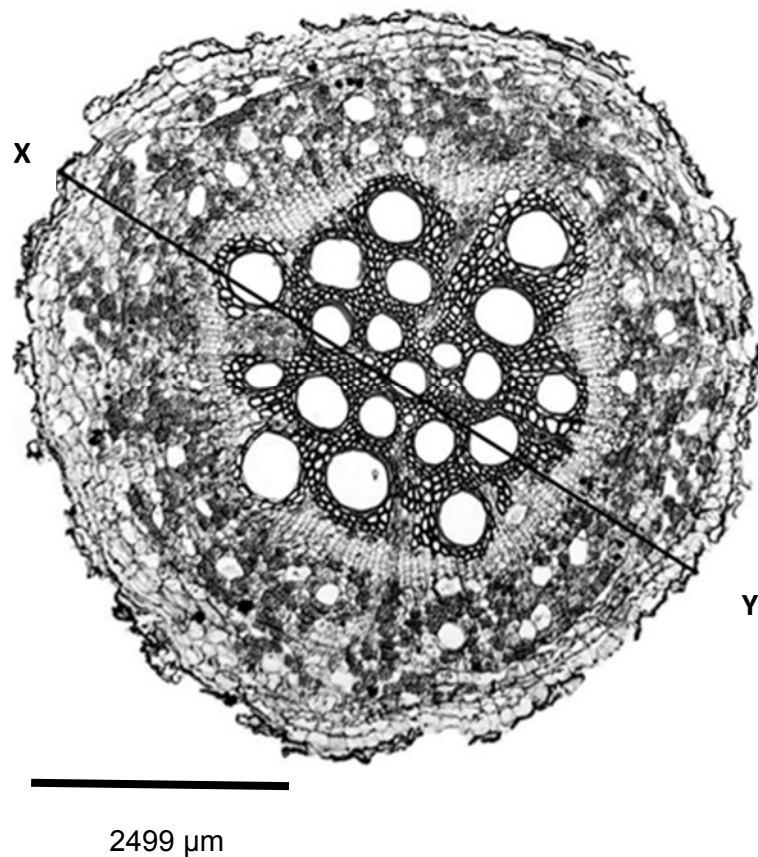


Fig. 3.2

In Fig. 3.2 the line X–Y is drawn across the diameter of the root.

Use the line X–Y and the scale bar to calculate the actual diameter of the root.

You may lose marks if you do not show your working or if you do not use appropriate units.

actual diameter = [3]

- (c) Observe the root on **M1** and the root in Fig. 3.2 and identify the differences between them. Record the observable differences in Table 3.1.

Table 3.1

feature	M1	Fig. 3.2

[2]

[Total: 13]